

Problem 45. Let a, b, c be distinct numbers such that the equations $x^2 + ax + 1 = 0$ and $x^2 + bx + c = 0$ have a common real root, and the equations $x^2 + x + a = 0$ and $x^2 + cx + b$ also have a common real root. Find the value of $a + b + c$.

Problem 46. Einstein chose a natural number $X \leq 100$, and Pythagoras is trying to guess this number. He can select two natural numbers M and N less than 100 and ask about $\gcd(X + M, N)$. Show that Pythagoras can determine Einstein's number with at most seven questions.

Problem 47. Let ABC be an acute-angled triangle with $AB < AC$, and let D be the foot of its altitude from A . Points B' and C' lie on the rays AB and AC , respectively, so that points B', C' and D are collinear and points B, C, B' and C' lie on one circle with centre O . Prove that if M is the midpoint of BC and H is the orthocentre of $\triangle ABC$, then $DHMO$ is a parallelogram.

Problem 48. Let A, B, C, a, b and c be positive integers. Suppose that the sum $S = A + B + C + a + b + c$ divides both $ABC + abc$ and $AB + BC + CA - ab - bc - ca$. Prove that S is composite.